

AARON BARTLETT

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OBJECTIVE

Computer Science graduate with background in Climate Modeling seeking opportunities in Environmental Data Science

EDUCATION

University of Michigan

Ann Arbor, MI

M.S. in Computer Science & Engineering

May 2026

B.S.E. in Computer Science & Engineering

December 2024

GPA: 3.84 / 4.00

Coursework: *Machine Learning, Database Management, Parallel Programming, Scalable Systems, Cybersecurity, Real-Time Computing, Operating Systems, Data Structures and Algorithms, Earth System Dynamics, Weather Systems*

WORK EXPERIENCE

Cooperative Institute for Great Lakes Research

Ann Arbor, MI

Atmospheric Science Research Specialist

January 2025 - Present

- Developed reproducible cloud workflows on NOAA's RDHPCS and PNNL's COMPASS to support the development of a two-way coupled Lake-Atmosphere Model for the Great Lakes by dynamically downscaling global climate data using a coupled WRF-FVCOM modeling framework
- Built and optimized real-time data pipelines for ingesting and processing eddy covariance observations across the Great Lakes, including automated QA/QC and gap-filling to ensure data quality and reliability
- Evaluated HRRR model performance against Great Lakes Evaporation Network observational data, conducting case studies and developing metrics to inform improvements to the Great Lakes Coastal Forecasting System

NOAA – National Weather Service

Silver Spring, MD

Web Development Intern

May 2023 – August 2023

- Designed an AWS-hosted scheduling service for Regional Science and Operations Officers to request and access forecast verification and model performance metrics, improving efficiency and scalability for regional meteorologists
- Created Python-based automation for performance charts and regional maps using WFS SQLite database data, significantly reducing manual processing and enhancing visualization of forecast verification results

RESEARCH & PROJECT EXPERIENCE

M-Bits Research Group

Ann Arbor, MI

Research Assistant

August 2024 – January 2025

- Researched fine-grain checkpointing strategies at the CUDA kernel level by analyzing PyTorch computational graphs and performance metrics using NVIDIA Nsight Profiler to optimize GPU memory usage and training efficiency

Statistics Online Computing Resource

Ann Arbor, MI

Data Methods Researcher

April 2024 – January 2025

- Derived data-driven personality types for U-M Nursing School client datasets of mice under observation using a multi-time-scale reinforcement learning framework with R, Python, SQL, and Tableau

Independent Projects

Renewable Grid Stress Modeling under Extreme Heat

- Simulated renewable power deficits during extreme heat events using Grid-Vantage with solar and wind forecasts

Augmented Reality Wildfire Simulation

- Built a sandbox-based AR visualization for fire mitigation education using an agent-based wildfire spread model

SKILLS

Coding Languages: Python, C/C++, SQL, R, CUDA, JavaScript, MATLAB, FORTRAN, Linux, GIS

ML & Statistics: Neural Networks, Agentic Models, Bayesian Models, Diffusion Models, TensorFlow, PyTorch

Data Engineering: Data Wrangling, QA/QC, NoSQL, PostgreSQL, MongoDB, Multi-Threading, Parallelization

ACTIVITIES

U-M Sun Radio Interferometer Space Experiment, *Software Developer*

December 2024 – May 2025

U-M EECS 445 Intro to Machine Learning, *Teaching Assistant*

August 2024 – Present

Michigan Ultimate Frisbee, *President/Captain*, *Ultiworld All-American*

June 2023 – Present